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WILLIAM M. GARDNER

education & training

Brigham and Women's Hospital, Harvard University

Residency, Internal Medicine, 2026 – 2029.

Columbia University

Doctor of Medicine, 2026.

University of Washington

MPH, Global Health Metrics, 2020.

Thesis: *Global, Regional, and National Estimates of Complete Coverage of Maternal Obstetric Care Interventions, 1990–2019.*

Harvard University

AB, Sociology, *Magna cum laude* with Highest Honors, 2017.

Thesis: *Institutions, Inequalities, and Infant Death: Understanding the Role of Health Care Decommodification in Latin America*

Minor: Global Health and Health Policy

experience

Institute for Health Metrics and Evaluation

Seattle, WA

External Contributor

2025 – present

Research Scientist

2020 – 2025

Post-Bachelor Fellow

2017 – 2020

Worked on the Neonatal and Child Health and Anemia teams as part of the Global Burden of Disease Study, modeling disease burden due to neonatal infectious diseases, population hemoglobin distributions and prevalence and causes of anemia, disease burden attributable to suboptimal breastfeeding, and coverage of a variety of maternal and child health interventions, including antenatal care, skilled birth attendance, facility births, cesarean section, and postnatal care.

Department of Sociology, Harvard University

Cambridge, MA

Research Assistant

2015 – 2017

Worked under the direction of Professor Jason Beckfield on a project researching the effects of income inequality on population health in Latin America using cross-national economic and health data from the World Income Inequality Database, World Bank, and World Values Survey.

Department of Global Health and Health Policy, Harvard University

Cambridge, MA

Cordeiro Summer Research Fellow

2016

Conducted research for individual project entitled "Evaluating Distributional Effects of the Welfare State on Health in Latin America," which culminated in my senior thesis; Duties included performing extensive literature reviews and cited reference searches, acquiring and cleaning relevant data sources, data modeling and statistical analysis, and manuscript writing.

publications

Journal articles

†Denotes equal contribution.

1. Martinez R, Thakur KT, Alviz LE, **Gardner WM**, Varela FJ, Hennis AJM, Oliveira e Souza R. The burden of noncommunicable and injury-related nervous system disorders in the Americas, 1990–2023: an analysis of the Global Burden of Disease Study 2023. *The Lancet Regional Health – Americas* 2026; 62.
2. Vongpradith A, Dominguez RMV, ..., **Gardner WM**, ..., Murray CJL, Kyu HH (GBD 2023 Diarrhoeal Disease and Enteric Infectious Diseases Collaborators). Global burden of enteric infectious diseases, diarrhoeal diseases, and corresponding aetiologies, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet Infectious Diseases* 2026.
3. Ledesma JR, Zhang M, ..., **Gardner WM**, ..., Murray CJL, Kyu HH (GBD 2023 TB HIV Collaborators). Global, regional, and national burden of tuberculosis and multidrug-resistant tuberculosis by HIV status, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet Infectious Diseases* 2026.
4. Nanditha NGA, Rhee DK, ..., **Gardner WM**, ..., Perumal N, Kassebaum NJ. Maternal and neonatal outcomes associated with haemoglobin concentration in pregnancy: a Burden of Proof study. *Nature Health* 2026; 1–14.
5. Teply CJ†, **Gardner WM**†, Noyes TS, McHugh TA, Tandiono HA, Nanditha NGA, Harris AA, Perumal N, Markt-Maloney J, Howatt BC, Taylor HJ, Humphrey K, Bhutta ZA, Girelli D, Hay SI, Murray CJL, Musallam KM, Nnodu O, Suchdev PS, Kassebaum NJ. Implications of changes in WHO haemoglobin elevation adjustment guidelines on global, regional, and national anaemia burden, 1990–2023: a population-based modelling study. *The Lancet Haematology* 2026; 13: e227–e240.
6. **Gardner WM**, Balte PP, Eckhardt CM, Morris JE, Bhatt SP, Couper DJ, Freedman ND, Jacobs DR, Kalhan R, Loehr LR, London SJ, Lutsey PL, Schwartz JE, White W, Yende S, Sanchez TR, Oelsner EC. Pipe and cigar use, lung function decline, and clinical outcomes: an analysis of the NHLBI Pooled Cohorts Study. *Thorax* 2026.
7. Khan Z, Jones AE, **Gardner WM**, McGroder CF, Murphy SO, Depender C, Mira-Sanchez M, Ogunlusi CO, Garcia CK, Wei Y, Baldwin MR. Inflammation, endothelial dysfunction, and frailty over 3 years in survivors of COVID-19 ARDS vs survivors of pneumonia. *CHEST Critical Care* 2026.
8. Hay SI, Ong KL, ..., **Gardner WM**, ..., Murray CJL, Gakidou E (GBD 2023 Disease and Injury and Risk Factor Collaborators). Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet* 2025; 406: 1873–1922.
9. Naghavi M, Kyu HH, ..., **Gardner WM**, ..., Wool EE, Murray CJL (GBD 2023 Causes of Death Collaborators). Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *The Lancet* 2025; 406: 1811–72.
10. Lee S, Son Y, ..., **Gardner WM**, ..., Yon DK, Kassebaum NJ (GBD 2021 Dietary Iron Deficiency Collaborators). Global, regional and national burden of dietary iron deficiency from 1990 to 2021: a Global Burden of Disease study. *Nature Medicine* 2025; 404: 2543–83.

11. Arndt MB, Abate YH, ..., **Gardner WM**, ..., Kassebaum NJ, Reiner RC (Global Nutrition Target Collaborators). [Global, regional, and national progress towards the 2030 global nutrition targets and forecasts to 2050: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 404: 2543–83.
12. Mokdad AH, Bisignano C, ..., **Gardner WM**, ..., Zyoud SH, Murray CJL (GBD 2021 US Burden of Disease Collaborators). [The burden of diseases, injuries, and risk factors by state in the USA, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 404: 2314–40.
13. Comfort H, McHugh TA, ..., **Gardner WM**, ..., Dandona R, Kassebaum NJ (GBD 2021 Global Stillbirths Collaborators). [Global, regional, and national stillbirths at 20 weeks' gestation or longer in 204 countries and territories, 1990–2021: findings from the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 404: 1955–88.
14. Brauer M, Roth GA, ..., **Gardner WM**, ..., Murray CJL, Gakidou E (GBD 2021 Risk Factors Collaborators). [Global burden and strength of evidence for 88 risk factors in 204 countries and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 403: 2162–203.
15. Ferrari AJ, Santomauro DF, ..., **Gardner WM**, ..., Vos T, Murray CJL (GBD 2021 Diseases and Injuries Collaborators). [Global incidence, prevalence, years lived with disability \(YLDs\), disability-adjusted life-years \(DALYs\), and healthy life expectancy \(HALE\) for 371 diseases and injuries in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 403: 2133–61.
16. Naghavi M, Ong KL, ..., **Gardner WM**, ..., Wool EE, Murray CJL (GBD 2021 Causes of Death Collaborators). [Global burden of 288 causes of death and life expectancy decomposition in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet* 2024; 403: 2100–32.
17. Steinmetz JD, Seeher KM, ..., **Gardner WM**, ..., Vos T, Dua T (GBD 2021 Nervous System Disorders Collaborators). [Global, regional, and national burden of disorders affecting the nervous system, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021](#). *The Lancet Neurology* 2024; 23: 344–81.
18. **Gardner WM**, Razo C, ..., Zoladl M, Kassebaum NJ (GBD 2021 Anaemia Collaborators). [Prevalence, years lived with disability, and trends in anaemia burden by severity and cause from 1990–2021: findings from the Global Burden of Disease Study 2021](#). *The Lancet Haematology* 2023; 10: e713–e734.
19. Thomson AM, McHugh TA, ..., **Gardner WM**, ..., Murray CJL, Kassebaum NJ (GBD 2021 Sickle Cell Disease Collaborators). [Global, regional, and national prevalence and mortality burden of sickle cell disease, 2000–2021: a systematic analysis from the Global Burden of Disease Study 2021](#). *The Lancet Haematology* 2023; 10: e585–e595.
20. Haakenstad A, Yearwood JA, ..., **Gardner WM**, ..., Murray CJL, Lozano R (GBD 2019 Healthcare Access and Quality Collaborators). [Assessing performance of the Healthcare Access and Quality Index, overall and by select age groups, for 204 countries and territories, 1990–2019: a systematic analysis from the Global Burden of Disease Study 2019](#). *The Lancet Global Health* 2022; 10: e1715–43.
21. Kyu HH, Sirota SB, ..., **Gardner WM**, ..., Brauer M, Hay SI (GBD 2019 LRI Collaborators). [Age-sex differences in the global burden of lower respiratory infections and risk factors, 1990–2019: results from the Global Burden of Disease Study 2019](#). *The Lancet Infectious Diseases* 2022; 22: 1626–47.

22. Gonçalves BP, Procter SR, Paul P, Chandna J, Lewin A, Seedat F, Koukounari A, Dangor Z, Leahy S, Santhanam S, John HB, Bramugy J, Bardaji A, Abubakar A, Nasambu C, Libster R, Yanotti CS, Horváth-Puhó E, Sørensen HT, Beek DVD, Bijlsma MW, **Gardner WM**, Kassebaum N, Trotter C, Bassat Q, Madhi SA, Lambach P, Jit M, Lawn JE. [Group B streptococcus infection during pregnancy and infancy: estimates of regional and global burden.](#) *The Lancet Global Health* 2022; 10: e807–19.
23. Fitzgerald R, Manguerra H, Arndt MB, **Gardner WM**, Chang YY, Zigler B, Taylor HJ, Bienhoff K, Smith DL, Murray CJL, Hay SI, Reiner RC, Kassebaum NJ. [Current dichotomous metrics obscure trends in severe and extreme child growth failure.](#) *Science Advances* 2021; 8.
24. Paulson KR, Kamath AM, ..., **Gardner WM**, ..., Wang H, Kassebaum NJ (GBD 2019 Under-5 Mortality Collaborators). [Global, regional, and national progress towards Sustainable Development Goal 3.2 for neonatal and child health: all-cause and cause-specific mortality findings from the Global Burden of Disease Study 2019.](#) *The Lancet* 2021; 398: 870–905.
25. Bhattacharjee NV, Schaeffer LE, ..., **Gardner WM**, ..., Zuniga YMH, Hay SI (Local Burden of Disease Exclusive Breastfeeding Collaborators). [Mapping inequalities in exclusive breastfeeding in low- and middle-income countries, 2000–2018.](#) *Nature Human Behaviour* 2021; 5: 1027–45.
26. Lozano R, Fullman N, ..., **Gardner WM**, ..., Zodpey S, Murray CJL (GBD 2019 Universal Health Coverage Collaborators). [Measuring universal health coverage based on an index of effective coverage of health services in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019.](#) *The Lancet* 2020; 396: 1250–84.
27. Murray CJL, Aravkin AY, ..., **Gardner WM**, ..., Afshin A, Lim SS (GBD 2019 Risk Factors Collaborators). [Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019.](#) *The Lancet* 2020; 396: 1223–49.
28. Vos T, Lim SS, ..., **Gardner WM**, ..., Naghavi M, Murray CJL (GBD 2019 Diseases and Injuries Collaborators). [Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019.](#) *The Lancet* 2020; 396: 1204–22.
29. Murray CJL, Abbafati C, ..., **Gardner WM**, ..., Vos T, Lim SS (GBD 2019 Viewpoint Collaborators). [Five insights from the Global Burden of Disease Study 2019.](#) *The Lancet* 2020; 396: 1135–59.
30. Dandona R, Kumar GA, ..., **Gardner WM**, ..., Sharma RS, Dandona L (India State-Level Disease Burden Initiative Child Mortality Collaborators). [Subnational mapping of under-5 and neonatal mortality trends in India: the Global Burden of Disease Study 2000–17.](#) *The Lancet* 2020; 395: 1640–58.
31. Troeger CE, Khalil IA, ..., **Gardner WM**, ..., Hay SI, Reiner RC (GBD 2017 Lower Respiratory Infections Collaborators). [Quantifying risks and interventions that have affected the burden of lower respiratory infections among children younger than 5 years: an analysis for the Global Burden of Disease Study 2017.](#) *The Lancet Infectious Diseases* 2020; 20: 60–79.
32. Troeger CE, Khalil IA, ..., **Gardner WM**, ..., Hay SI, Reiner RC (GBD 2017 Diarrhoeal Disease Collaborators). [Quantifying risks and interventions that have affected the burden of diarrhoea among children younger than 5 years: an analysis of the Global Burden of Disease Study 2017.](#) *The Lancet Infectious Diseases* 2020; 20: 37–59.

33. Swaminathan S, Hemalatha R, ..., **Gardner WM**, ..., Toteja GS, Dandona L (India State-Level Disease Burden Initiative Malnutrition Collaborators). [The burden of child and maternal malnutrition and trends in its indicators in the states of India: the Global Burden of Disease Study 1990–2017](#). *The Lancet Child & Adolescent Health* 2019; 3: 855–70.
34. Bhattacharjee NV, Schaeffer LE, ..., **Gardner WM**, ..., Dwyer-Lindgren L, Hay SI. [Mapping exclusive breastfeeding in Africa between 2000 and 2017](#). *Nature Medicine* 2019; 25: 1205–12.
35. Lozano R, Fullman N, ..., **Gardner WM**, ..., Lim SS, Murray CJL (GBD 2017 SDG Collaborators). [Measuring progress from 1990 to 2017 and projecting attainment to 2030 of the health-related Sustainable Development Goals for 195 countries and territories: a systematic analysis for the Global Burden of Disease Study 2017](#). *The Lancet* 2018; 392: 2091–138.
36. Stanaway JD, Afshin A, ..., **Gardner WM**, ..., Zodpey S, Murray CJL (GBD 2017 Risk Factor Collaborators). [Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017](#). *The Lancet* 2018; 392: 1923–94.
37. James SL, Abate D, ..., **Gardner WM**, ..., Vos T, Murray CJL (GBD 2017 Disease and Injury Incidence and Prevalence Collaborators). [Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017](#). *The Lancet* 2018; 392: 1789–858.
38. Kyu HH, Abate D, ..., **Gardner WM**, ..., Zuhlke LJ, Murray CJL (GBD 2017 DALYs and HALE Collaborators). [Global, regional, and national disability-adjusted life-years \(DALYs\) for 359 diseases and injuries and healthy life expectancy \(HALE\) for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017](#). *The Lancet* 2018; 392: 1859–922.
39. Roth GA, Abate D, ..., **Gardner WM**, ..., Naghavi M, Murray CJL (GBD 2017 Causes of Death Collaborators). [Global, regional, and national age-sex-specific mortality for 282 causes of death in 195 countries and territories, 1980–2017: a systematic analysis for the Global Burden of Disease Study 2017](#). *The Lancet* 2018; 392: 1736–88.

Other media

1. **William Gardner**, Nicholas Kassebaum, Theresa McHugh. [Anemia afflicts nearly 1 in 4 people worldwide, but there are practical strategies for reducing it](#). *The Conversation*, 2023.
2. **William Gardner**. [A recent study found that nearly 1 in 4 people worldwide are anemic](#). *KCBS Radio*, 2023.

conference presentations

1. Angelova Y, Kwak HR, Zimmerman P, Thearle M, **Gardner WM**, Mirza O, Cheung S. Risk Factors for Discharge Against Medical Advice of Patients with Opioid Use Disorder. Academy of Consultation-Liaison Psychiatry Annual Conference, 2025.
2. Kwak HR, Zimmerman P, Cheung S, Thearle M, Mirza O, **Gardner WM**, Angelova Y. Discharges against medical advice in patients with opioid use disorder in an urban community hospital. New York County Psychiatric Society Poster Competition, 2025.

3. Mira-Sanchez M, Jones AE, Khan Z, **Gardner WM**, McGroder C, Depender C, Murphy SO, Garcia CK, Roh DJ, Baldwin MR. [Subclinical hypercoagulable viscoelastic characteristics as-sociate with post-acute sequelae of SARS-CoV-2 physical symptoms in COVID Acute lung in-jury/acute respiratory distress syndrome survivors](#). American Thoracic Society Annual Conference, 2025.
4. Teply CJ, **Gardner WM**, Tandiono HA, Noyes TS, Nanditha NGA, Taylor HJ, Howatt BC, Perumal N, Kassebaum NJ. [Evaluation of Hemoglobin Elevation Adjustment Method and Its Effect On Estimating the Global Burden of Anemia](#). Society for Pediatric and Perinatal Epidemiology Annual Conference, 2024.
5. **Gardner WM** and Kassebaum NJ. [Global, Regional, and National Prevalence of Anemia and Its Causes in 204 Countries and Territories, 1990–2019](#). American Society for Nutrition Annual Conference, 2020. (Finalist for Emerging Leaders in Nutrition Science Award)
6. **Gardner WM** and Kassebaum NJ. [Global, Regional, and National Prevalence and Trends in Infant Breastfeeding Status in 204 Countries and Territories, 1990–2019](#). American Society for Nutrition Annual Conference, 2020.
7. **Gardner WM** and Kassebaum NJ. [Global, Regional, and National Coverage of Maternal Care Interventions](#). Society for Pediatric and Perinatal Epidemiology Annual Conference, 2020.

**invited
presentations**

William Gardner 2023, 2024, 2025. "Anemia As a Global Public Health Challenge."
Institute of Human Nutrition, Columbia University, New York, NY.

teaching

Intro to Point-of-Care Ultrasound: eFAST Exam, Columbia University VP&S (Instructor)	2025
Point-of-Care Ultrasound, Ready4Residency, Columbia University VP&S (TA)	2025

**professional
organizations**

Society for Pediatric and Perinatal Epidemiology	2019 – present
American Society for Nutrition	2019 – present
WHO scientific advisory group for <i>Group B Streptococcus</i> vaccine value proposition	2019 – 2022

**peer
review**

Nature

**honors
& awards**

Alpha Omega Alpha Medical Honor Society (Columbia University VP&S)	2026
Michael H. Aranow Memorial Prize (Columbia University VP&S)	2026
Dr. Charles E. Hamilton Award (Columbia University VP&S)	2026
Albert B. Knapp Scholarship (Columbia University VP&S)	2026
Fellow, International Fellows Program (Columbia University SIPA)	2025 – 2026
Margaret and Charles Clark '43 Memorial Scholarship (Columbia University VP&S)	2022 – 2026
Future GIM Scholar, Mid-Atlantic Region of the Society of General Internal Medicine	2023
T35 Training Grant, National Heart, Lung, and Blood Institute (Columbia University)	2023
Emerging Leaders in Nutrition Science Award, American Society for Nutrition (finalist)	2020
Phi Beta Kappa	2017
David McCord Prize (Harvard University)	2017
John Harvard Scholar	2016 – 2017
Cordeiro Health Policy Summer Research Fellowship	2016